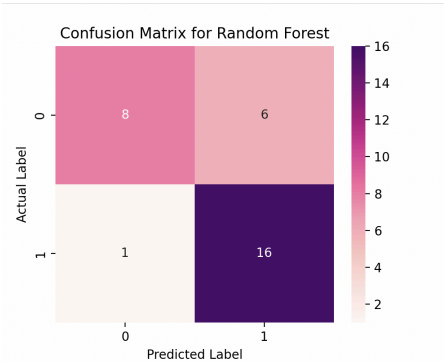
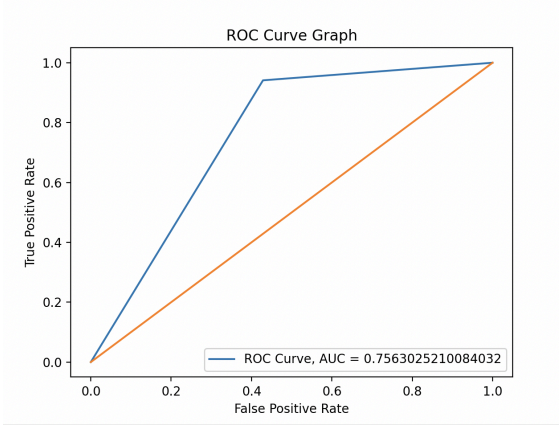
Final Model Decisions

- Selected Model: Logistic Regression
- Reason: Logistic Regression model had the highest accuracy, 77.4%. The Logistic Regression model also had a high ROC-AUC curve score. Below is the Confusion Matrix, alongside the ROC-AUC curve.
- Overview: The model performs binary classification, and is evaluated using ROC-AUC.
- 



Front-end Inputs

- Forecasted Weather; a string; Rain, Cold, Warm, Clear Skies.
- Mood Level Before Going Out (Throughout the Day); a string; Stressed, Sad, Neutral, Relaxed, Happy.
- Energy Level Before Going Out (Throughout the Day); a string; Exhausted, Low Energy, Neutral, Energized, Extremely Energized.
- Responsibility Level Before Going Out (Throughout the Day); a string; Low (No deadlines in the next week), Medium (Deadlines in the next week), High (Deadlines in the next 3 days).

Input Preprocessing:

- weather is one-hot encoded.
- mood is ordinal encoded:
    - Stressed -> 0
    - Sad -> 1
    - Neutral -> 2
    - Relaxed -> 3
    - Happy -> 4
- energy is ordinal encoded:
    - Exhausted -> 0
    - Low Energy -> 1

    - Neutral -> 2
    - Energized -> 3
    - Extremely Energized -> 4
- responsibility is ordinal encoded:
    - Low (No deadlines in the next week) -> 0
    - Medium (No deadlines in the next week) -> 1
    - High (No deadlines in the next week) -> 2
- mood_x_responsibility is created

Weather Category Mapping (using Open-Meteo)

- Clear Skies: 0, 1
- Warm: 2, 3, 45, 48
- Rain: 51, 53, 55, 56, 57, 61, 63, 65, 66, 67, 71, 73, 75, 77, 80, 81, 82, 85, 86, 95, 96, 99
- Cold: if temp < 55 F -> “Cold”
- * The model only accepts these 4 categories.

Final Feature Order:

- 'Forecasted Weather_Clear Skies', 'Forecasted Weather_Cold', 'Forecasted Weather_Rain', 'Forecasted Weather_Warm', 'mood_ord', 'energy_ord', 'responsibility_ord', 'mood_x_responsibility'

Front-end Request

```
{
  "forecasted_weather": "Clear Skies",
  "mood_level": "Happy",
  "energy_level": "Energized",
  "responsibility_level": "Low (No deadlines in the next week)"
}
```

Example Backend Output

```
{
"will_go_out": true,
"prediction": 1,
"probability": 0.82
}
```

- will_go_out is the final prediction. Boolean. True = Yes. False = No.
- prediction is an integer (0 or 1)
- probability is a float (range: 0 to 1)